



MORIAH COLLEGE

Year 12

MATHEMATICS

Extension 2

Date: **Wednesday 8th August, 2001**

Time Allowed: 3 hours, plus 5 minutes reading time.

Examiners: J. Taylor

Instructions:

- Attempt ALL questions.
- ALL questions are of equal value.
- All necessary working should be shown in every question.
Marks may be deducted for careless or badly arranged work.
- Standard integrals are printed on page 6
- Board approved calculators may be used.
- Answer each question in a SEPARATE writing booklet.
- You may ask for extra Writing Booklets, if you need them.

Question 1 (15 marks)

a) Find

$$\text{i) } \int \frac{\cos^{-1} \frac{2x}{3}}{\sqrt{9-4x^2}} dx \quad 2$$

$$\text{ii) } \int x^2 \tan^{-1} \frac{x}{2} dx \quad 3$$

$$\text{iii) } \int \frac{dx}{\sqrt{2x^2+3x}} \quad 4$$

b) If 6

$$I_n = \int_0^1 x^n (1-x)^{1/2} dx \quad (n > 0)$$

prove that

$$I_n = \left(\frac{2n}{2n+3} \right) I_{n-1}$$

and evaluate I_2 .

Question 2 (15 marks)

- a) Sketch the region in the Argand Plane consisting of those points z for which

$$|z + 3 - i| > 4 \text{ intersects with } -\frac{3\pi}{8} < \arg z \leq -\frac{3\pi}{4} \quad 3$$

- b) Given that

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

- i) Find, in terms of π , an approximations for $e^{i\pi}$ in the form $x + iy$ using the above four terms of the series. 1

- ii) Use your calculator where necessary to plot this approximation on an Argand diagram 2

- iii) If z is any complex number, prove that both 4

$$z^n + \bar{z}^n$$

and

$$e^z + e^{\bar{z}}$$

are pure real numbers

- c) i) Find the five fifth roots of unity. 2

- ii) If $\omega = \text{cis} \frac{2\pi}{5}$, show that $1 + \omega + \omega^2 + \omega^3 + \omega^4 = 0$ 1

- iii) Show that $z_1 = \omega + \omega^4$ and $z_2 = \omega^2 + \omega^3$ are the roots of the equation 2
- $$z^2 + z - 1 = 0$$

Question 3 (15 marks)

a) Sketch $y = \sqrt{\cos 3x}$ in the domain: $-\pi \leq x \leq \pi$ 2

b) The rational function $f(x)$ is defined

$$f(x) = \frac{x - p}{(x - q)(x - r)}$$

- i) Write $f(x)$ as the sum of two partial fractions. 2
- ii) If p lies between q and r , use a neat sketch to explain why $f(x)$ can assume all real values. 2
- iii) If p does not lie between q and r , use a neat sketch to explain why $f(x)$ *cannot* assume all real values. 2
- iv) Show that in either case, the gradient of the tangent at the midpoint of the interval between q and r is independent of p 3

c) If x, y are positive integers such that $x - y > 1$, then prove that

$$x! + y! > (x - 1)! + (y + 1)! \quad \text{3}$$

Question 4 (15 marks)

- a) For the ellipse $\frac{y^2}{50} + \frac{x^2}{32} = 1$, find
- i) the eccentricity, 2
 - ii) the coordinates of the foci S and S' . 1
- b) Explain why $\frac{x^2}{\lambda - 23} + \frac{y^2}{5 - \lambda} = 1$ cannot represent the equation of an ellipse. 1
- c) Normals to the ellipse $4x^2 + 9y^2 = 36$ at points $P(3\cos\alpha, 2\sin\alpha)$ and $Q(3\cos\beta, 2\sin\beta)$ are at right angles to each other. Show that
- i) the gradient of the normal at P is $\frac{3\sin\alpha}{2\cos\alpha}$, 2
 - ii) $4\cot\alpha\cot\beta = -9$. 2
- d) $P\left(5p, \frac{5}{p}\right), p > 0$ and $Q\left(5q, \frac{5}{q}\right), q > 0$ are two points on the hyperbola, $H, xy = 25$.
- i) Derive the equation of the chord PQ , 2
 - ii) State the equations of the tangents at P and Q , 1
 - iii) If the tangents at P and Q intersect at R , find the co-ordinates of R . 2
 - iv) If the secant PQ passes through the point $S(15, 0)$, find the locus of R . 2

Question 5 (15 marks)

- a) When $x^3 - kx^2 - 10kx + 25$ is divided by $x - 2$ the remainder is 9. Find the value of k . 2
- b) A polynomial function is $P(x) = x^5 + x^4 + 13x^3 + 13x^2 - 48x - 48$. Factorise $P(x)$ over the field of
- i) real numbers, 2
- ii) complex numbers. 1
- c) The equation $x^5 - 5x^4 - x^3 + 3x^2 + 1 = 0$ has roots $\alpha, \beta, \gamma, \delta$. Find the equations with roots
- i) $\frac{1}{\alpha} + 2, \frac{1}{\beta} + 2, \frac{1}{\gamma} + 2, \frac{1}{\delta} + 2$, 2
- ii) $\alpha^2 - 1, \beta^2 - 1, \gamma^2 - 1, \delta^2 - 1$ 3
- d) $\phi(x)$ is a polynomial of degree 5 such that $\phi(x) - 1$ is divisible by $(x - 1)^3$ and $\phi(x)$ itself is divisible by x^3 . Derive an expression for $\phi(x)$. 5

Question 6 (15 marks)

- a) i) Prove that $2\pi \int_1^5 x\sqrt{4-(x-3)^2} dx = 12\pi^2$

3

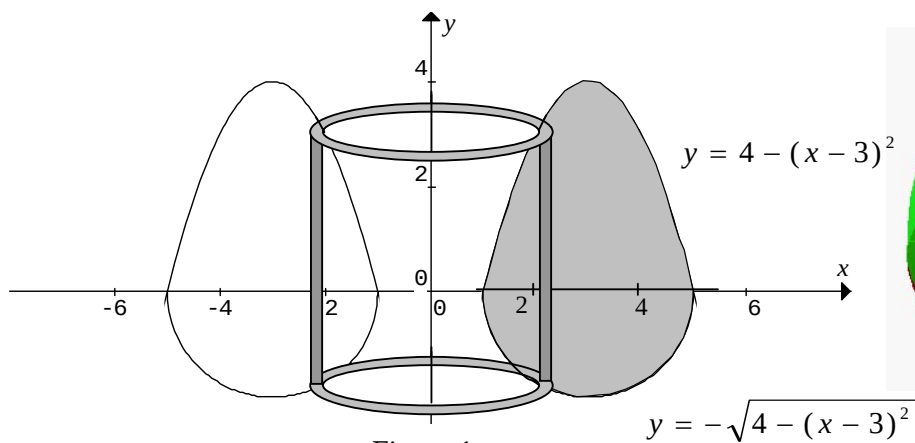


Figure 1

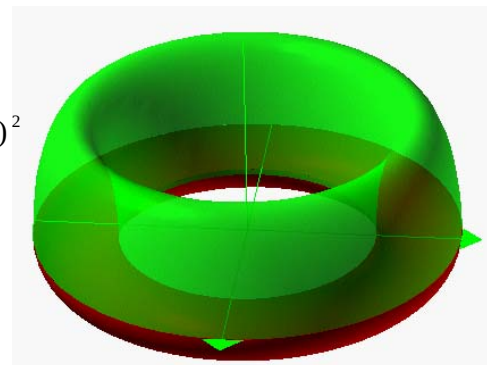


Figure 2

- ii) The solid in fig.2 is formed by rotating about the y -axis the area bounded by the parabola $y = 4 - (x - 3)^2$ and the semi-circle $y = -\sqrt{4 - (x - 3)^2}$.

Use the method of cylindrical shells to calculate the volume generated.

3

Question 6 b) is on the next page.

Question 6

- b) A circle in a horizontal x - y plane has centre O , radius R units. At each point $P(x, y)$ on this circle, another circle is constructed perpendicular to the original circle in the plane containing the radius at that point. The radius r of such a circle (see the shaded circle, figure 1) is given by
- $$r = xy.$$

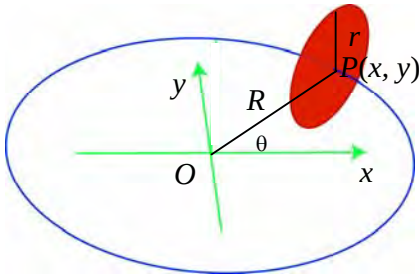


Figure 1

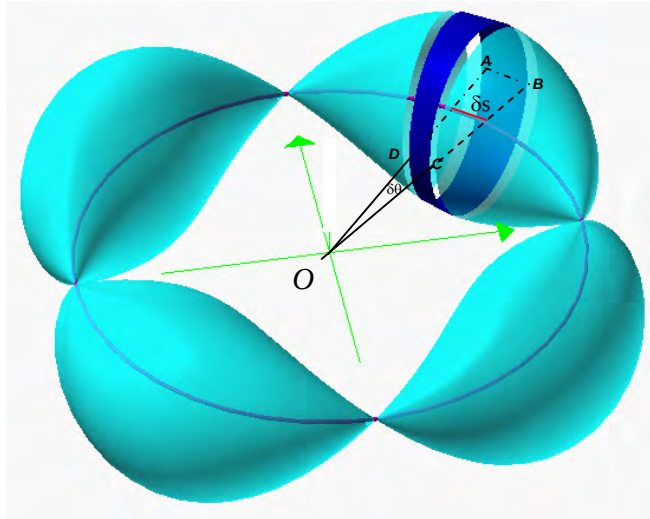


Figure 2

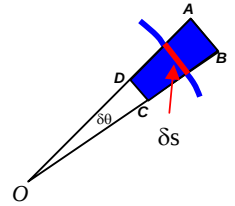


Figure 3

- i) If $\angle POx = \theta$, prove that $r = \frac{R^2 \sin 2\theta}{2}$

2

As P moves around the horizontal circle, the vertical circles will form a surface drawn in figure 2.

A section is taken in the first quadrant by slicing the figure with two vertical planes from the centre of the horizontal circle. This section can be approximated by the wedge-shaped solid in figure 4:

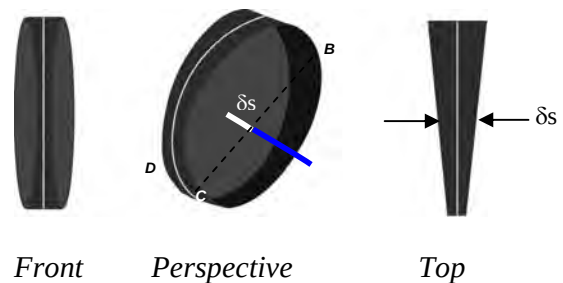


Figure 4

- ii) If the thickness of the section at the centre is the straight line length δs , explain *briefly* why the volume δV of the section in figure 4 is $\delta V = \pi r^2 \delta s$.
- iii) If $\delta \theta$ is the angle between the radii used to slice the figure (see figure 3), prove that
- a) $\delta V \approx R \pi^2 \delta \theta$
- b)
$$V = \frac{R^5}{4} \pi \lim_{\delta \theta \rightarrow 0} \sum_{\theta=0}^{2\pi} \sin^2 2\theta \delta \theta$$
- iv) Find the volume of the solid.

1

3

3

Question 7 (15 marks)

- a) A vertical pole PO of height l units is standing on a flat plane. Suspended from the top P of the pole is a string, also of length l , at the end of which is attached a particle of mass m . The pole begins to rotate so that the mass describes a horizontal circle of radius r units with a uniform angular velocity so that the centre of the circle is h units below P .

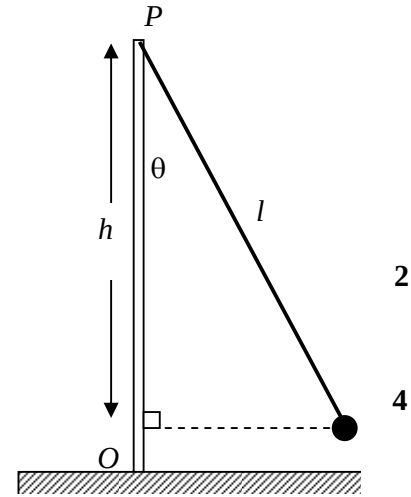
The string makes an angle θ with the pole.

- i) If the speed of the particle is v , prove that

$$\tan \theta = \frac{v^2}{gr}$$

- ii) When the particle is uniformly rotating at a height $h = \frac{l}{2}$ the string suddenly snaps, and the particle travels freely through the air to land at a point Q .

Find the distance OQ in terms of l .



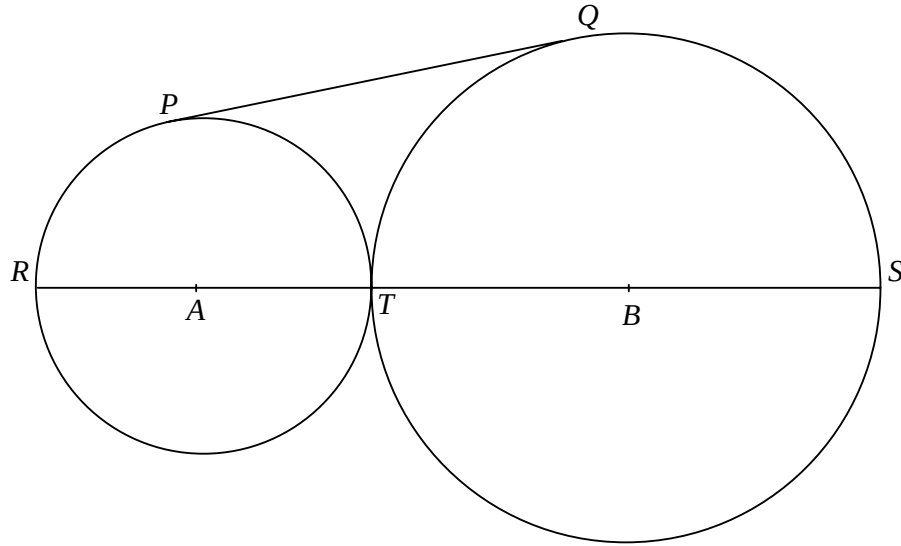
- b) The only force acting on a particle moving in a straight line is a resistance $m\lambda(c + v)$ acting in the same line. The mass of the particle is m , its velocity is v , and λ and c are positive constants. The particle starts to move with velocity u (>0) and comes to rest in time T . At time $\frac{1}{2}T$ its velocity is $\frac{1}{4}u$. Show that

i) $c = \frac{1}{8}u$,

ii) at time t , $8\frac{v}{u} = 9e^{-\lambda t} - 1$.

Question 8 (15 marks)

a)



Copy the above diagram into your answer booklet.

In the diagram, circles with centres A and B touch each other at T . PQ is a direct common tangent. The line of centres cuts the circles at R and S as shown.

RP and SQ produced meet at X .

Prove that $\angle RXS$ is a right angle.

5

b) A sequence of polynomials (called the *Bernoulli Polynomials*) is defined inductively by the three conditions:

$$1) \quad B_0(x) = 1$$

$$2) \quad B'_n(x) = nB_{n-1}(x)$$

$$3) \quad \int_0^1 B_n(x) dx = 0 \text{ if } n \geq 1$$

i) Prove that $B_1(x) = x - \frac{1}{2}$

3

ii) Prove that if $B_n(x+1) - B_n(x) = nx^{n-1}$ and

$$g(x) = B_{n+1}(x+1) - B_{n+1}(x)$$

then

$$g'(x) = (n+1)nx^{n-1}$$

and deduce an expression for $g(x)$.

4

iii) Prove by induction that

$$B_n(x+1) - B_n(x) = nx^{n-1} \text{ if } n \geq 1.$$

3

End of Paper.

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0.$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0.$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0.$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a \neq 0, \quad -a \leq x \leq a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left\{ x + \sqrt{(x^2 - a^2)} \right\}, \quad |x| > |a|$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left\{ x + \sqrt{(x^2 + a^2)} \right\}$$

NOTE: $\ln x = \log_e x, \quad x > 0.$

Question 1 (15 marks)

a) i)

$$\text{Let } u = \cos^{-1} \frac{2x}{3}$$

$$du = \frac{-2}{\sqrt{9-4x^2}} dx$$

$$I = \int -\frac{u}{2} du \quad 2$$

$$= -\frac{u^2}{4} + c$$

$$= -\frac{1}{4} \left(\cos^{-1} \frac{2x}{3} \right)^2 + c$$

ii)

$$\int x^2 \tan^{-1} \frac{x}{2} dx \quad v' = x^2 \quad u = \tan^{-1} \frac{x}{2}$$

$$v = \frac{x^3}{3} \quad u' = \frac{2}{4+x^2}$$

$$\int uv' dx = uv - \int vu' dx$$

$$= \frac{x^3}{3} \tan^{-1} \frac{x}{2} - \frac{2}{3} \int \frac{x^3}{4+x^2} dx \quad 1$$

$$= \frac{x^3}{3} \tan^{-1} \frac{x}{2} - \frac{2}{3} \int \frac{4x + x^3 - 4x}{4+x^2} dx \quad 1$$

$$= \frac{x^3}{3} \tan^{-1} \frac{x}{2} - \frac{2}{3} \int \left(x - \frac{4x}{4+x^2} \right) dx$$

$$= \frac{x^3}{3} \tan^{-1} \frac{x}{2} - \frac{x^2}{3} + \frac{4}{3} \ln(4+x^2) + c \quad 2$$

iii)

$$I = \frac{1}{\sqrt{2}} \int \frac{dx}{\sqrt{x^2 + \frac{3x}{2}}}$$

$$= \frac{1}{\sqrt{2}} \int \frac{dx}{\sqrt{\left(x + \frac{3}{4}\right)^2 - \frac{9}{16}}} \quad 2$$

$$= \frac{1}{\sqrt{2}} \ln \left[x + \frac{3}{4} + \sqrt{x^2 + \frac{3x}{2}} \right] + C$$

1b)

$$\begin{aligned}
I_n &= \int_0^1 x^n (1-x)^{\frac{1}{2}} dx \\
&= \left[-\frac{2}{3} x^n (1-x)^{\frac{3}{2}} \right]_0^1 + \frac{2n}{3} \int_0^1 x^{n-1} (1-x)^{\frac{3}{2}} dx \\
&= \frac{2n}{3} \int_0^1 x^{n-1} (1-x)(1-x)^{\frac{1}{2}} dx \\
&= \frac{2n}{3} \int_0^1 x^{n-1} (1-x)^{\frac{1}{2}} - x^n (1-x)^{\frac{1}{2}} dx \\
&= \frac{2n}{3} [I_{n-1} - I_n] \\
I_n + \frac{2n}{3} I_n &= \frac{2n}{3} I_{n-1} \\
I_n \left(\frac{2n+3}{3} \right) &= \left(\frac{2n}{3} \right) I_{n-1} \\
I_n &= \left(\frac{2n}{2n+3} \right) I_{n-1}
\end{aligned}$$

4

$$\begin{aligned}
I_1 &= 0 + \frac{2}{3} \int_0^1 (1-x)^{\frac{3}{2}} dx \\
&= \frac{2}{3} \times -\frac{2}{5} \left[(1-x)^{\frac{5}{2}} \right]_0^1 \\
&= \frac{4}{15}
\end{aligned}$$

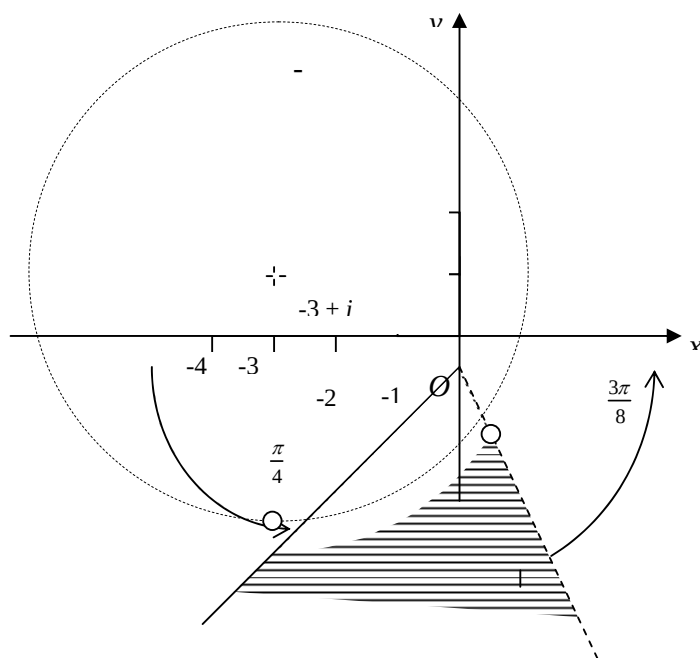
Hence

$$\begin{aligned}
I_2 &= \frac{4}{7} I_1 \\
&= \frac{4}{7} \times \frac{4}{15} \\
I_2 &= \frac{16}{105}
\end{aligned}$$

3

Question 2 (15 marks)

a)



3

c) i)

$e^{i\pi} \approx 1 + i\pi + \frac{(i\pi)^2}{2!} + \frac{(i\pi)^3}{3!}$ $= 1 + i\pi - \frac{\pi^2}{2} - \frac{i\pi^3}{6}$ $= 1 - \frac{\pi^2}{2} + i\left(\pi - \frac{\pi^3}{6}\right)$ (1)	$e^{i\pi} \approx 1 - \frac{\pi^2}{2} + i\left(\pi - \frac{\pi^3}{6}\right)$ $\approx -3.9438 + i(-2.0261)$ (1)
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ii) a) Let $z = r(\cos \theta + i \sin \theta)$. Then

$$\bar{z} = r(\cos \theta - i \sin \theta)$$

$$= r(\cos(-\theta) + i \sin(-\theta))$$

Hence, by de Moivre's Theorem,

$$z^n + \bar{z}^n = r^n(\cos n\theta + i \sin n\theta) + r^n(\cos(-n\theta) + i \sin(-n\theta))$$

$$= r^n(\cos n\theta + i \sin n\theta + \cos(n\theta) - i \sin(n\theta)) \quad (2)$$

$$= 2r^n \cos n\theta$$

which is totally real

b) Similarly

$$e^z + e^{\bar{z}} = 1 + z + \frac{z^2}{2!} + \frac{z^3}{3!} + \dots + 1 + \bar{z} + \frac{\bar{z}^2}{2!} + \frac{\bar{z}^3}{3!} + \dots \quad (2)$$

$$= 2 + (z + \bar{z}) + (z^2 + \bar{z}^2) + (z^3 + \bar{z}^3) + \dots$$

which is totally real from above.

b) i) $z^5 = 1$

$$(\cos \theta + i \sin \theta)^5 = 1$$

$$\cos 5\theta + i \sin 5\theta = 1$$

$$\cos 5\theta = 1 \text{ and } \sin 5\theta = 0$$

$$5\theta = 0, 2\pi, 4\pi, 6\pi, 8\pi$$

$$\theta = 0, \frac{2\pi}{5}, \frac{4\pi}{5}, \frac{6\pi}{5}, \frac{8\pi}{5}$$

$$z = \text{cis } \frac{2k\pi}{5} \quad k = 0, 1, 2, 3, 4$$

1

1

ii) If $\omega = \text{cis } \frac{2\pi}{5}$

$$\omega^2 = \text{cis } \frac{4\pi}{5}, \quad \omega^3 = \text{cis } \frac{6\pi}{5}, \quad \omega^4 = \text{cis } \frac{8\pi}{5}$$

$$1 + \omega + \omega^2 + \omega^3 + \omega^4 = \text{sum of roots} \\ = 0$$

1

iii) $z_1 = \omega + \omega^4$ and $z_2 = \omega^2 + \omega^3$

$$z_1 + z_2 = \omega + \omega^4 + \omega^2 + \omega^3 = -1$$

1

$$z_1 z_2 = (\omega + \omega^4)(\omega^2 + \omega^3)$$

$$= \omega^3 + \omega^4 + \omega^6 + \omega^7$$

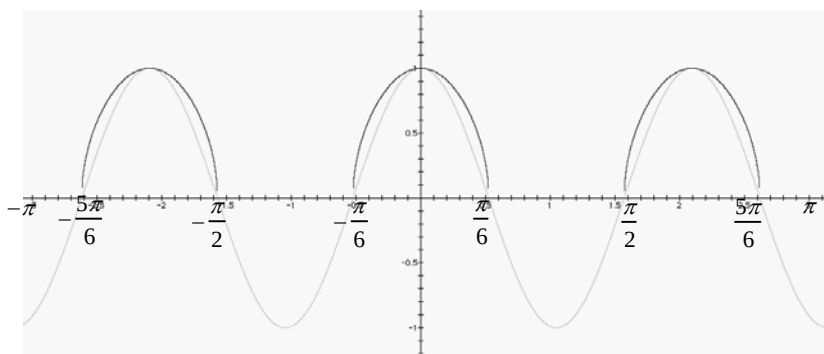
$$= \omega^3 + \omega^4 + \omega + \omega^2$$

$$= -1$$

1

Question 3 (15 marks)

a) $y = \cos 3x$  $y = \sqrt{\cos 3x}$ 



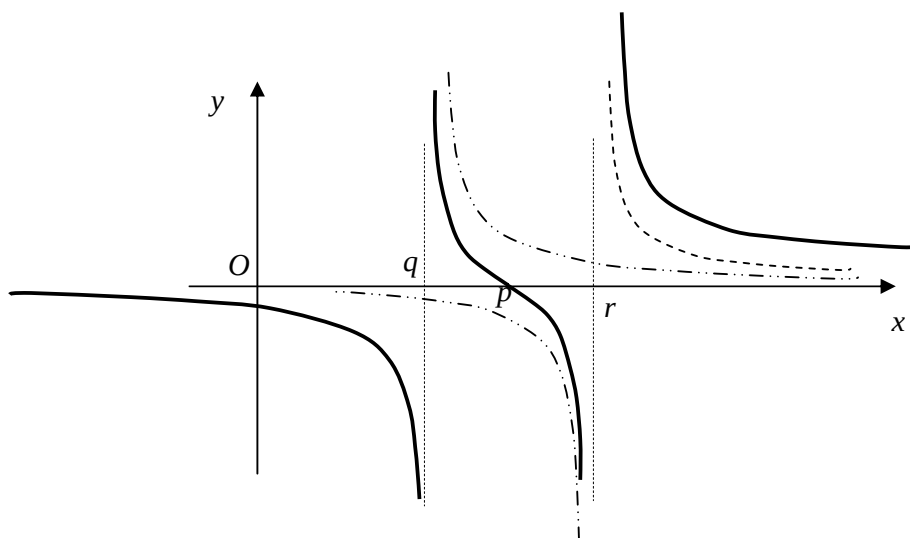
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b) i)

$$\begin{aligned} \frac{x-p}{(x-q)(x-r)} &= \frac{A}{(x-q)} + \frac{B}{(x-r)} \\ &= \frac{\frac{q-p}{q-r}}{(x-q)} + \frac{\frac{r-p}{r-q}}{(x-r)} \quad (\text{Heaviside}) \\ &= \frac{\frac{q-p}{q-r}}{(x-q)} - \frac{\frac{q-p}{q-r}}{(x-r)} \end{aligned}$$

2

ii) The partial fractions show us the graph of the function can be viewed as the sum of two hyperbolas, both of which are defined for all values in the interval $q < x < r$. The sum of these graphs can only be zero in the given interval if one branch is above the x axis, and one is below the x axis, as in the diagram below. This shows that $f(x)$ approaches infinity in both directions and is defined for all x in that interval. Hence $f(x)$ assumes all real values in that interval.



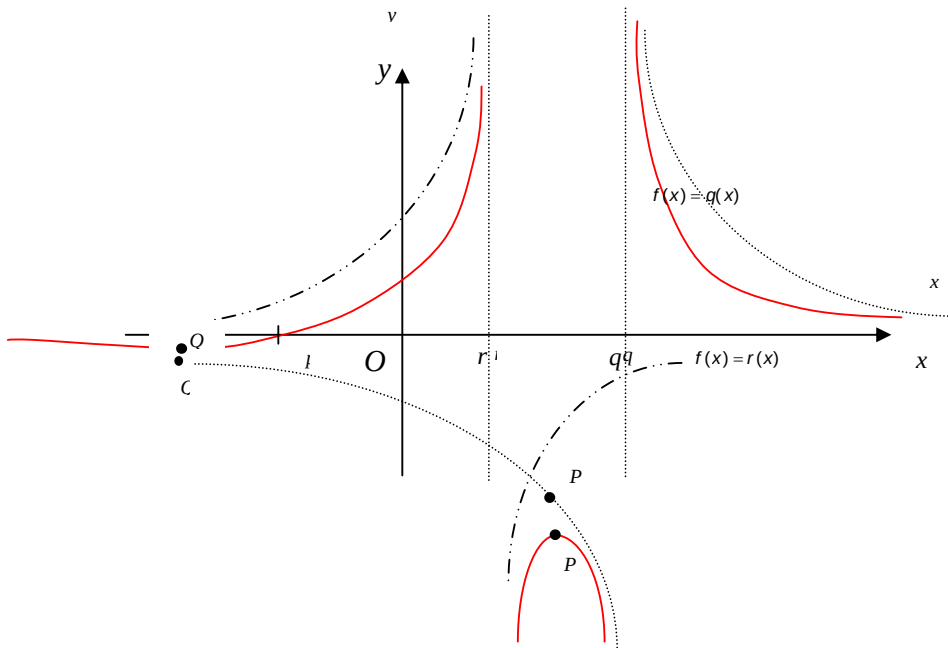
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ii) From part i) we have

$$\begin{aligned}
 f(x) &= \frac{q-p}{q-r} + \frac{r-p}{q-r} \\
 &= \frac{L}{(x-q)} + \frac{M}{(x-r)} \\
 &= q(x) + r(x)
 \end{aligned}$$

where L, M are positive. if $p < r < q$

Let P be a point on $y = q(x)$, $r < x < q$. Both $q(x)$ and $r(x)$ are negative, and so the point P' on $f(x)$ is lower than P . In a similar way, Q on $q(x)$ (where $x < r$) will be lower than Q' on $f(x)$ since $r(x) > 0$ and $q(x) < 0$ here. But *there will always be a vertical gap between P and Q* . Hence there will always be a gap between Q' and P' and the result follows.



The relative positions of q and r simply produce reflections of the above idea.

2

$$\begin{aligned}
f'(x) &= \frac{-\frac{q-p}{q-r}}{(x-q)^2} + \frac{-\frac{r-p}{r-q}}{(x-r)^2} \\
&= \frac{p-q}{(q-r)(x-q)^2} + \frac{r-p}{(q-r)(x-r)^2} \\
&= \frac{p-q}{(q-r)(x-q)^2} + \frac{r-p}{(q-r)(x-r)^2} \\
f'\left(\frac{q+r}{2}\right) &= \frac{p-q}{(q-r)\left(\frac{r-q}{2}\right)^2} + \frac{r-p}{(q-r)\left(\frac{q-r}{2}\right)^2} \\
&= \frac{4}{(q-r)^3} [p-q+r-p] \\
&= -\frac{4}{(q-r)^2}
\end{aligned}$$

3

which is independent of p

c) Consider $x! + y! - [(x-1)! + (y+1)!] = (x-1)![x-1] + y![1 - (y+1)]$ 2

$$\begin{aligned}
&= (x-1)![x-1] - y![y] \\
&> y![y] - y![y] \\
&= 0
\end{aligned}$$

$$\therefore x! + y! - [(x-1)! + (y+1)!] > 0$$

$$\therefore x! + y! - [(x-1)! + (y+1)!] > [(x-1)! + (y+1)!]$$

Question 4 (15 marks)

- a) i) Since $50 > 32$, the major axis is the y axis and so

$$a^2 = 50 - 3\sqrt{2} \quad b^2 = 32$$

$$a = 5\sqrt{2} \quad b = 4\sqrt{2}$$

$$b^2 = a^2(1 - e^2)$$

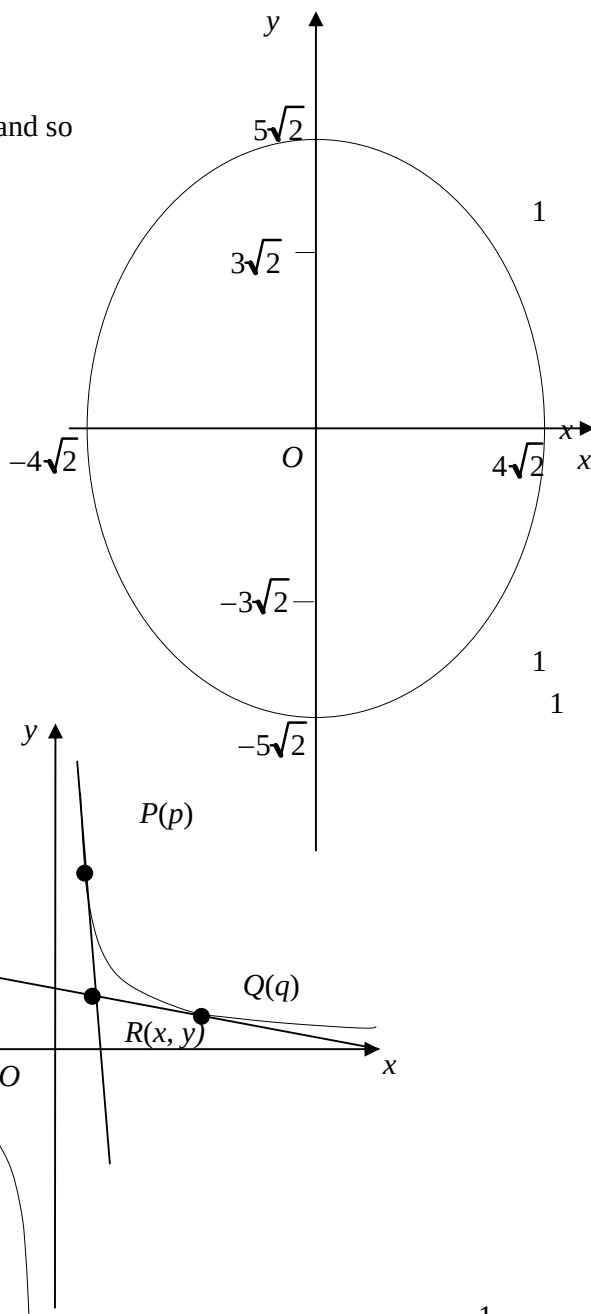
$$e^2 = \frac{a^2 - b^2}{a^2}$$

$$= 1 - \frac{32}{50}$$

$$= \frac{18}{50}$$

$$e = \frac{3}{5}$$

ii) $S(0, \pm ae) = S(0, \pm 3\sqrt{2})$



- b) For an ellipse
 $\lambda - 23 > 0$ and $5 - \lambda > 0$
 i.e. $\lambda > 23$ and $\lambda < 5$, which is not possible.

c) $4x^2 + 9y^2 = 36$

i)

$$8x + 18y \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -\frac{4x}{9y}$$

At $P(3\cos\alpha, 2\sin\alpha)$, $\frac{dy}{dx} = -\frac{2\cos\alpha}{3\sin\alpha}$, $m_{\text{normal}} = \frac{3\sin\alpha}{2\cos\alpha}$ ($m_1 m_2 = -1$)

At $Q(3\cos\beta, 2\sin\beta)$, $\frac{dy}{dx} = -\frac{2\cos\beta}{3\sin\beta}$, $m_{\text{normal}} = \frac{3\sin\beta}{2\cos\beta}$ ($m_1 m_2 = -1$)

ii)

$$\frac{3\sin\alpha}{2\cos\alpha} \cdot \frac{3\sin\beta}{2\cos\beta} = -1$$

$$\frac{4\cos\alpha\cos\beta}{9\sin\alpha\sin\beta} = -1$$

$$4\cot\alpha\cot\beta = -9$$

2

d) .i)

$$m_{PQ} = \frac{\frac{5}{p} - \frac{5}{q}}{5p - 5q}$$

$$= -\frac{1}{pq}$$

1

$$PQ: \quad y - \frac{5}{p} = -\frac{1}{pq}(x - 5p)$$

$$x + pqy = 5(p + q)$$

1

ii) Tangent at P is $x + p^2y = 10p$

Tangent at Q is $x + q^2y = 10q$

1

iii) $R: (p^2 - q^2)y = 10(p - q)$

$$y = \frac{10}{p + q}$$

1

$$x + \frac{10p^2}{p + q} = 10p$$

$$x = 10p - \frac{10p^2}{p + q}$$

1

$$= \frac{10pq}{p + q}$$

iv) $S(15, 0)$

$$15 = 5(p + q)$$

$$p + q = 3$$

1

$$x = \frac{10pq}{3} \quad y = \frac{10}{3}$$

$$\text{Locus of } R \text{ is } y = \frac{10}{3}$$

1

Question 5 (15 marks)

a) $P(x) = x^3 - kx^2 - 10kx + 25$ $P(2) = 9$
 $8 - 4k - 20k + 25 = 9$ 1
 $-16k = -16$ 1
 $k = 1$

b) $P(x) = x^5 + x^4 + 13x^3 + 13x^2 - 48x - 48$. Factorise $P(x)$ over the field of
i)

$$\begin{aligned} P(x) &= x^4(x+1) + 13x^2(x+1) - 48(x+1) \\ &= (x+1)(x^4 + 13x^2 - 48) \\ &= (x+1)(x^2 + 16)(x^2 - 3) \\ &= (x+1)(x^2 + 16)(x - \sqrt{3})(x + \sqrt{3}) \end{aligned}$$

1
1

ii) $P(x) = (x+1)(x+4i)(x-4i)(x-\sqrt{3})(x+\sqrt{3})$ 1

c) $x^5 - 5x^4 - x^3 + 3x^2 + 1 = 0$ has roots $\alpha, \beta, \gamma, \delta$.

i) The new equation has roots $x = \frac{1}{\alpha} + 2$ etc.

$$\alpha = \frac{1}{x-2}$$

Equation is $\left(\frac{1}{x-2}\right)^5 - 5\left(\frac{1}{x-2}\right)^4 - \left(\frac{1}{x-2}\right)^3 + 3\left(\frac{1}{x-2}\right)^2 + 1 = 0$ 1

$$1 - 5(x-2) - (x-2)^2 + 3(x-2)^3 + (x-2)^5 = 0 \quad (\text{full marks at this point})$$
 1

$$x^5 - 15x^4 + 53x^3 - 99x^2 + 115x - 49 = 0$$

ii) The new equation has roots $x = \alpha^2 - 1$ etc.

$$\alpha = \sqrt{x+1}$$

Equation is $(\sqrt{x+1})^5 - 5(\sqrt{x+1})^4 - (\sqrt{x+1})^3 + 3(\sqrt{x+1})^2 + 1 = 0$ 1

$$(x+1)^2 \sqrt{x+1} - 5(x+1)^2 - (x+1)\sqrt{x+1} + 3\sqrt{x+1} + 1 = 0$$
 1

$$[(x+1)^2 - (x+1)]\sqrt{x+1} = 5x^2 + 7x + 1$$
 1

$$(x^2 + x)\sqrt{x+1} = 5x^2 + 7x + 1$$

(full marks at this point) 1

$$(x^2 + x)^2 (x+1) = (5x^2 + 7x + 1)^2$$

$$x^5 - 22x^4 - 67x^3 - 58x^2 - 14x - 1 = 0$$

d) $\phi(x) = x^3(ax^2 + bx + c)$

$$\phi(x) - 1 = (x-1)^3(ax^2 + dx + e) \quad 1$$

$$x^3(ax^2 + bx + c) - 1 = (x-1)^3(ax^2 + dx + e)$$

$$ax^5 + bx^4 + cx^3 - 1 = (x^3 - x^2 + x - 1)(ax^2 + dx + e) \quad 1$$

$$= ax^5 + (d - 3a)x^4 + (3a - 3d + e)x^3 + (3d - a - 3e)x^2 + (3e - d)x - e$$

$$a = a$$

$$b = d - 3a$$

$$c = 3a - 3d + e \quad 1$$

$$3d - a - 3e = 0$$

$$3e - d = 0$$

$$-e = -1$$

Solving we get $e = 1, \quad d = 3, \quad a = 6, \quad c = 10. \quad b = -15$ 1

$$\therefore \phi(x) = x^3(6x^2 - 15x + 10) \quad 1$$

$$= 6x^5 - 15x^4 + 10x^3$$

Alternately,

$$\phi(x) = ax^5 + bx^4 + cx^3 \quad 1$$

$$P(x) = ax^5 + bx^4 + cx^3 - 1$$

Since $P(x)$ is divisible by $(x-1)^3$, $x=1$ is a triple zero of $P(x)$.

$$P'(x) = 5ax^4 + 4bx^3 + 3cx^2 \quad 1$$

$$P''(x) = 20ax^3 + 12bx^2 + 6cx$$

$$P''(1) = 5a + 4b + 3c = 0$$

$$P'(1) = 20a + 12b + 6c = 0 \quad 1$$

$$P(1) = a + b + c - 1 = 0$$

Solving simultaneously, $a = 6, \quad b = -15, \quad c = 10$ 1

$$\phi(x) = 6x^5 - 15x^4 + 10x^3 \quad 1$$

Question 6 (15 marks)

Q6a)

i)

$$\begin{aligned}\int_1^5 x\sqrt{4-(3-x)^2} dx &= \int_{-2}^2 (3+u)\sqrt{4-u^2} du \\ &= \int_{-2}^2 3\sqrt{4-u^2} du + \int_{-2}^2 u\sqrt{4-u^2} du \\ &= 3 \times \text{semicircle} + \text{integral of odd function} \\ &= 3 \times \frac{1}{2} \pi \times 2^2 + 0 \\ &= 6\pi\end{aligned}$$

4

$$2\pi \int_1^5 x\sqrt{4-(3-x)^2} dx = 12\pi^2$$

ii)

b)

$$\delta V = 2\pi r h \delta x$$

$$\begin{aligned}V &= 2\pi \int_1^5 xy dx \\ &= 2\pi \int_1^5 x \left[4 - (x-3)^2 + \sqrt{4-(3-x)^2} \right] dx \\ &= 2\pi \int_1^5 x \left[4 - (x-3)^2 \right] dx + 2\pi \int_1^5 x \left[\sqrt{4-(3-x)^2} \right] dx \\ &= 2\pi \int_1^5 -x^3 + 6x^2 - 5x dx + 12\pi^2 \\ &= 64\pi + 12\pi^2\end{aligned}$$

4

The volume is $64\pi + 12\pi^2$ units³

b

i)

$$r = xy$$

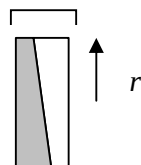
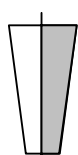
$$= R \cos \theta R \sin \theta$$

$$= R^2 \cos \theta \sin \theta$$

$$= \frac{R^2}{2} \sin 2\theta$$

2

ii)



δV = cylinder with base radius r , height δs

$$= \pi r^2 \delta s$$

1

iii) a)

$$\text{Arc length} = R \delta \theta$$

$$\delta s \approx R \delta \theta$$

$$\delta V \approx \pi r^2 R \delta \theta$$

$$= R \pi r^2 \delta \theta$$

1

b)

$$\begin{aligned}
V &= \lim_{\delta\theta \rightarrow 0} \sum_{\delta\theta=0}^{2\pi} R\pi r^2 \delta\theta \\
&= R\pi \lim_{\delta\theta \rightarrow 0} \sum_{\delta\theta=0}^{2\pi} r^2 \delta\theta \\
&= R\pi \lim_{\delta\theta \rightarrow 0} \sum_{\delta\theta=0}^{2\pi} \frac{R^4}{4} \sin^2 2\theta \delta\theta \\
&= \frac{R^5}{4} \pi \lim_{\delta\theta \rightarrow 0} \sum_{\delta\theta=0}^{2\pi} \sin^2 2\theta \delta\theta \\
V &= \frac{\pi R^5}{4} \int_0^{2\pi} \sin^2 2\theta d\theta \\
&= \frac{\pi R^5}{8} \int_0^{2\pi} 1 - \cos 4\theta d\theta \\
&= \frac{\pi R^5}{8} \left[\theta - \frac{\sin 4\theta}{4} \right]_0^{2\pi} \\
&= \frac{\pi^2 R^5}{4}
\end{aligned}$$

1

2

Question 7 (15 marks)

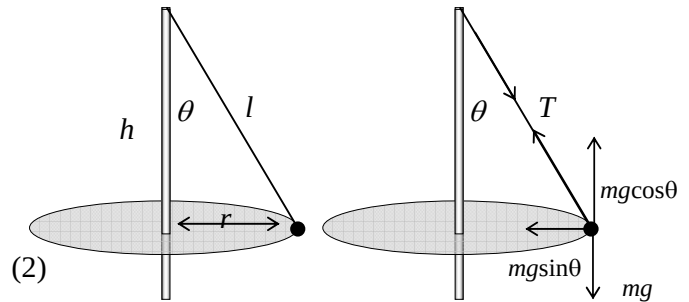
Solution

i) Let the tension in the string be T . Then

$$T \sin \theta = m \frac{v^2}{r} \quad (1)$$

$$T \cos \theta = mg \quad (2)$$

Dividing (1) by (2) gives $\tan \theta = \frac{v^2}{gr}$



ii) When the string breaks, $\theta = 60^\circ$ and so

$$h = \frac{l}{2}, \quad r = \frac{l}{2\sqrt{3}}, \quad \tan \theta = \sqrt{3}, \quad v = r \sqrt{\frac{g}{h}} = \sqrt{\frac{3l \cdot g}{2}}$$

When the string breaks, the particle will travel *horizontally* at a *tangent* to the circle of motion.

Taking an origin at the point of breaking, the x and y coordinates will be

$$x = vt, \quad y = -\frac{1}{2}gt^2$$

$$y = -\frac{1}{2}g \frac{x^2}{v^2}$$

This gives

$$y = -\frac{1}{2}g \frac{x^2}{v^2}$$

and so when $y = -l/2$, the distance R travelled before landing will be

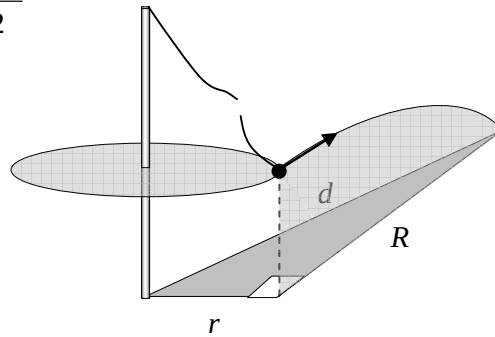
$$\begin{aligned} R^2 &= \frac{l}{2g} \cdot 2v^2 \\ &= lv^2 \\ &= \frac{3l^2}{2} \end{aligned}$$

But the distance required is d , hence

$$\begin{aligned} d^2 &= r^2 + R^2 \\ &= \frac{3l^2}{4} + \frac{3l^2}{2} \\ &= \frac{9l^2}{4} \end{aligned}$$

Hence the particle will land a distance $\frac{3l}{2}$ m from

O(2)



$$m \frac{dv}{dt} = -m\lambda(c+v)$$

$$\frac{dv}{dt} = -\lambda(c+v)$$

$$\frac{dt}{dv} = -\frac{1}{\lambda(c+v)}$$

$$\int_0^t dt = -\frac{1}{\lambda} \int_u^v \frac{dv}{c+v}$$

$$t = -\frac{1}{\lambda} \log\left(\frac{c+v}{c+u}\right)$$

$$= \frac{1}{\lambda} \log\left(\frac{c+u}{c+v}\right)$$

$$\text{At } T = T, v = 0 \text{ and at } t = \frac{T}{2}, v = \frac{u}{4}$$

$$T = \frac{1}{\lambda} \log\left(\frac{c+u}{c}\right), \quad \frac{T}{2} = \frac{1}{\lambda} \log\left(\frac{c+u}{c+\frac{u}{4}}\right) \quad (4)$$

$$\frac{1}{\lambda} \ln\left(\frac{c+u}{c}\right) = \frac{2}{\lambda} \ln\left(\frac{c+u}{c+\frac{u}{4}}\right)$$

$$\left(\frac{c+u}{c}\right) = \left(\frac{c+u}{c+\frac{u}{4}}\right)^2 \quad 1$$

$$\left(c+\frac{u}{4}\right)^2 = c(c+u)$$

$$c^2 + \frac{cu}{2} + \frac{u^2}{16} = c^2 + cu$$

$$\frac{cu}{2} = \frac{u^2}{16} \quad 1$$

$$c = \frac{u}{8}$$

ii)

$$t = \frac{1}{\lambda} \ln\left(\frac{c+u}{c+v}\right)$$

$$\lambda t = \ln\left(\frac{\frac{u}{8}+u}{\frac{u}{8}+v}\right) \quad 1$$

$$e^{\lambda t} = \frac{9u}{u+8v} \qquad 1$$

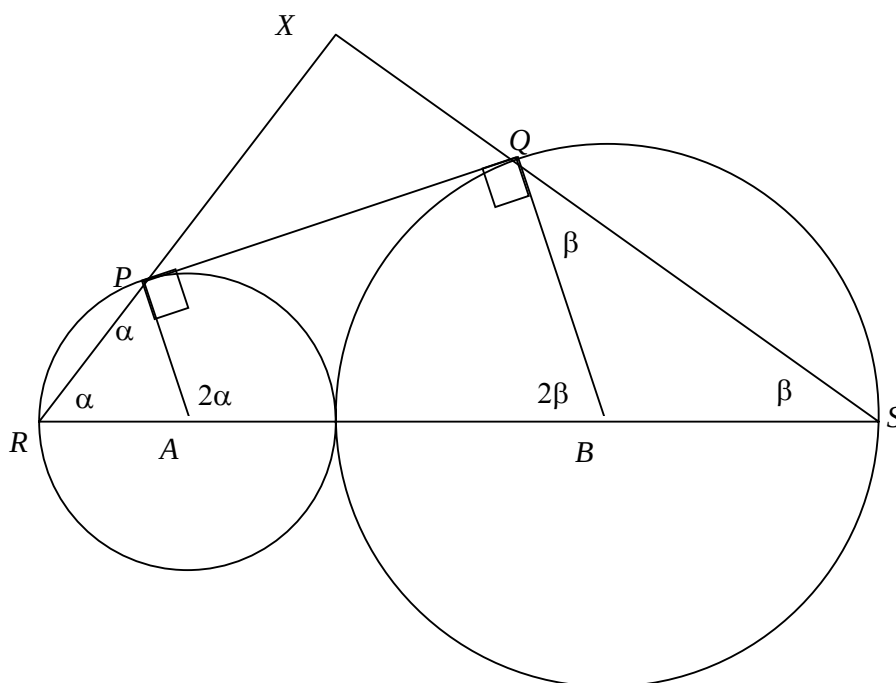
$$\frac{u+8v}{9u} = e^{-\lambda t}$$

$$1 + \frac{8v}{u} = 9e^{-\lambda t}$$

$$\frac{8v}{u} = 9e^{-\lambda t} - 1 \qquad 1$$

Question 8 (15 marks)

a)



Construction:

Produce RP and SQ to X . Join PA and QB . Label angles as marked.

$$\angle ARP = \angle APR = \alpha \quad (\angle\text{s of isos } \Delta, \text{ equal radii})$$

$$\angle BSQ = \angle BQS = \beta \quad (\angle\text{s of isos } \Delta, \text{ equal radii}) \quad 1$$

$$\angle PAB = 2\alpha \quad (\text{Exterior } \angle\text{s of } \Delta = \text{sum of interior opp } \angle\text{s}) \quad 1$$

$$\angle QBA = 2\beta \quad (\text{Exterior } \angle\text{s of } \Delta = \text{sum of interior opp } \angle\text{s})$$

$$2\alpha + 2\beta = 180^\circ \quad (\text{Cointerior } \angle\text{s of } \parallel \text{ lines supplementary, } PA \parallel QS)$$

$$\therefore \alpha + \beta = 90^\circ$$

$$\therefore \alpha + \beta + \theta = 180^\circ \quad (\angle \text{ sum of } \Delta = 180^\circ) \quad 1$$

$$\therefore \theta = 90^\circ \quad 1$$

$$\text{i.e. } \angle RXS = 90^\circ$$

b)

i)

$$B_1'(x) = 1 \cdot B_0(x)$$

$$= 1$$

$$B_1(x) = x + C$$

To evaluate C :

$$\int_0^1 B_1(x) dx = \int_0^1 (x + C) dx$$

$$0 = \int_0^1 x dx + C \quad \left[\text{since } \int_0^1 C dx = C \right]$$

$$C = -\int_0^1 x dx \quad 3$$

$$= -\left[\frac{x^2}{2} \right]_0^1$$

$$= -\frac{1}{2}$$

$$\therefore B_1(x) = x - \frac{1}{2}$$

ii)

$$g(x) = B_{n+1}(x+1) - B_{n+1}(x)$$

$$g'(x) = B'_{n+1}(x+1) - B'_{n+1}(x)$$

$$= (n+1)B_n(x+1) - (n+1)B_n(x) \quad 2$$

$$= (n+1)[B_n(x+1) - B_n(x)]$$

$$= (n+1)nx^{n-1} \quad \text{by assumption.}$$

$$\therefore g(x) = (n+1)x^n + C$$

iii)

$$B_1(x+1) - B_1(x) = \left(x+1 - \frac{1}{2}\right) - \left(x - \frac{1}{2}\right)$$

$$= 1 \quad 5$$

$$= 1x^0$$

and the proposition is true for $n = 1$.

Assuming the result is true for some positive integer $n = k$, we now consider the function

$$g(x) = B_{k+1}(x+1) - B_{k+1}(x)$$

$$g'(x) = B'_{k+1}(x+1) - B'_{k+1}(x)$$

$$= (k+1)B_k(x+1) - (k+1)B_k(x)$$

$$= (k+1)[B_k(x+1) - B_k(x)]$$

$$= (k+1)kx^{k-1} \quad \text{by assumption.}$$

$$\therefore g(x) = (k+1)x^k + C$$

Hence

$$B_{k+1}(x+1) - B_{k+1}(x) = (k+1)x^k + C$$

$$B_{k+1}(1) - B_{k+1}(0) = (k+1).0 + C$$

$$C = 0$$

and so $B_{n+1}(x+1) - B_{n+1}(x) = (n+1)x^n$, and the proposition is true for $n+1$ if it is true for n .

But the proposition is true for $n=1$. Hence, by mathematical induction,

$$B_n(x+1) - B_n(x) = nx^{n-1} \quad \text{if } n \geq 1. \quad (3)$$